

Fitness Buddy

AI-Powered Agentic Health & Fitness Coach

Problem Statement

Millions of people struggle to maintain a healthy lifestyle because of busy schedules, lack of personalized guidance, inconsistent motivation, and limited access to affordable fitness experts. Many fitness applications provide generic workout plans that do not consider a user's age, fitness level, dietary preferences, lifestyle, or medical limitations. Fitness Buddy addresses these challenges by delivering intelligent, personalized, and affordable AI-powered fitness coaching.

The Challenge

Develop an AI-powered Fitness Buddy using IBM watsonx.ai, IBM Granite Models, IBM Cloud and IBM BOB that provides personalized workouts, Indian nutrition guidance, healthy lifestyle recommendations, progress tracking, and continuous motivation through an intelligent conversational AI.

Project Objective

Build an AI virtual fitness coach capable of generating customized home and gym workouts, Indian meal plans, BMI, BMR, TDEE, calorie analysis, hydration advice, progress monitoring, and healthy habit recommendations based on each user's profile and goals.

Key Features

- AI Fitness Chat Assistant
- Personalized Workout Planner
- Home & Gym Workout Generator
- Indian Diet Planner
- BMI, BMR & TDEE Calculator
- Water Intake Calculator
- Progress Dashboard
- Exercise Library
- Daily Motivation
- Weekly & Monthly Plans

Role of Agentic AI

Agentic AI understands user goals, collects health information, reasons over fitness requirements, creates personalized recommendations, adapts plans based on progress, and provides continuous guidance while prioritizing user safety.

Technology Stack

- IBM watsonx.ai
- IBM Granite Foundation Models
- IBM Cloud

- IBM BOB
- Python Flask
- HTML, CSS, JavaScript, Bootstrap 5
- SQLite / MongoDB

Novelty and Uniqueness

Fitness Buddy combines AI coaching, Indian nutrition planning, intelligent workout generation, progress analytics, and personalized motivation into a single platform specifically designed for Indian users.

Future Scope

- Voice-enabled Fitness Coach
- AI Posture Detection
- Smartwatch Integration
- Family Health Profiles
- Wearable Device Synchronization
- Regional Language Support

Expected Outcome

The system enables users to achieve healthier lifestyles through personalized, evidence-based fitness guidance while making professional coaching accessible, affordable, and available 24/7.

Vision

To empower every individual with an intelligent AI fitness coach that delivers personalized workouts, Indian nutrition guidance, and sustainable healthy lifestyle recommendations.